CHAPTER 12

ATOMS

CHAPTER ESSENCE AT A GLANCE:
Rutherford's alpha scattering experiment, Bohr's quantized hydrogen model, spectral series, and atomic energy level transitions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Bohr's Stationary Orbits:** Electrons revolve only in discrete non-radiating stationary orbits without losing energy continuously through electromagnetic radiation.
- **Angular Momentum Quantization:** An electron can only occupy stable orbits where orbital angular momentum is an integral multiple of Planck's constant divided by two pi.
- **Energy Levels & Transitions:** Radiation is emitted or absorbed only when an electron jumps discontinuously between stationary states with discrete photon energy.
- **Hydrogen Spectral Series:** Transitions cascading to lower quantized levels produce distinct spectral line series; transitions ending at the second orbit form the visible Balmer series.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} = 0.529\, n^2 \text{ AA}$$

Bohr radius of hydrogen atom in state n

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

Total energy of electron in nth Bohr orbit

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Rydberg formula for hydrogen spectral series

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Balmer series is the ONLY hydrogen spectral series that falls in the VISIBLE region of electromagnetic spectrum.
- **Alert:** [Valuation Rubric] State all three Bohr postulates with exact mathematical formulas .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Alpha-particle scattering experiment (Geiger-Marsden experiment): observations and conclusions [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Geiger-Marsden Experiment Setup]** Alpha particles from a radioactive Bismuth-214 source collimated into a narrow beam strike a thin gold foil (thickness ~2.1 × 10^{-7} m). Scattered alpha particles are detected on a rotatable zinc sulfide (ZnS) screen using a microscope. [1 Mark]

• **[Key Observations]** 1. More than 99% of alpha particles pass straight through the gold foil with little or no deflection (< 1°). 2. About 1 in 8000 alpha particles is deflected through large angles (> 90°). 3. Very rarely, an alpha particle rebounds back along its original path (deflection ~180°). [1 Mark]

• **[Conclusions]** 1. Most of the space inside an atom is empty. 2. The entire positive charge and almost the entire mass of the atom is concentrated in a tiny central region called the NUCLEUS (radius ~10^{-15} m vs atom radius ~10^{-10} m). [1 Mark]

• **Key Exam Result:** Rutherford discovered atomic nucleus: atom is mostly empty space with tiny massive positive center

2. Distance of closest approach formula r 0 = $\frac{2kZe^2}{K}$ and nuclear size estimate [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Principle: Energy Conservation]** Alpha particle (charge q_1 = 2e, mass m) approaches head-on towards a target nucleus of atomic number Z (charge q_2 = Z e). At distance of closest approach r_0, the initial kinetic energy K is completely converted into electrostatic potential energy U (alpha particle stops momentarily before rebounding): [1.5 Marks]

• **[Derivation of r_0]** K = U = (one-fourth (0.25)$\pi\varepsilon_0$) · (q_1 q_2 divided by r_0) = (one-fourth (0.25)$\pi\varepsilon_0$) · [(2e)(Z e) divided by r_0] = (one-fourth (0.25)$\pi\varepsilon_0$) · (2 Z e^2 divided by r_0). Distance of closest approach: r_0 = (one-fourth (0.25)$\pi\varepsilon_0$) · (2 Z e^2 divided by K). For 7.7 MeV alpha particles on gold (Z = 79): r_0 approx. (9 × 10^9 × 2 × 79 × (1.6 × 10^{-19})^2) divided by (7.7 × 10^6 × 1.6 × 10^{-19}) approx. 3.0 × 10^{-14} m = 30 fm. Sets an upper limit on nuclear size! [1.5 Marks]

• **Key Exam Result:** Distance of closest approach r_0 = (one-fourth (0.25)$\pi\varepsilon_0$) · (2 Z e^2 divided by K)

3. Impact parameter and scattering angle relation (qualitative) [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Impact Parameter Definition (b)]** The perpendicular distance of the initial velocity vector of the alpha particle from the central line of the nucleus when the particle is far away. [1 Mark]

• **[Relation with Scattering Angle θ]** Formula: b = (one-fourth (0.25)$\pi\varepsilon_0$) · [(Z e^2 cot(θ divided by 2)) divided by K]. Physical deduction: 1. Large impact parameter (b >> nuclear radius): Electrostatic repulsive force is weak; alpha particle suffers very small or zero deflection (θ approx. 0°). 2. Zero impact parameter (b = 0): Head-on collision; cot(θ divided by 2) = 0 implies θ divided by 2 = 90° implies θ = 180° (particle rebounds directly backward). [1 Mark]

• **Key Exam Result:** b = (one-fourth (0.25)$\pi\varepsilon_0$) [Z e^2 cot(θ divided by 2) divided by K]; b=0 corresponds to θ=180°

4. Rutherford's planetary model of atom and its two major limitations (stability and discrete spectrum) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Rutherford's Model Overview]** Atom consists of a central positive massive nucleus orbited by negatively charged electrons, analogous to planets orbiting the Sun. [1 Mark]

• **[Two Major Limitations]** 1. Atomic Instability: According to classical Maxwellian electromagnetic theory, an accelerating charge (electron orbiting in circular path has centripetal acceleration a = v^2 divided by r) must continuously radiate energy in the form of EM waves. As it loses energy, its orbital radius must shrink continuously, spiraling into the nucleus in ~10^{-8} seconds. In reality, atoms are stable! 2. Discrete Line Spectra: Classical collapse would emit a continuous spectrum of all frequencies. In reality, atoms emit discrete, sharp spectral lines characteristic of the element. [2 Marks]

• **Key Exam Result:** Classical model fails: predicts nuclear collapse in 10^{-8} s and continuous spectra

5. Rydberg formula for hydrogen spectrum 1 divided by \lambda = R(1 divided by n 1^2 - 1 divided by n 2^2) [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Rydberg Formula Statement]** The wavenumber (1 divided by λ) of emitted radiation during transition between levels n_2 and n_1 ($n_2 > n_1$) is: ν bar = 1 divided by $\lambda = R_H \cdot Z^2$ [1 divided by n_1^2 - 1 divided by n_2^2]. [1 Mark]

• **[Rydberg Constant (R H)]** From Bohr's energy expression: $R_H = (m e^4)$ divided by $(8 \varepsilon_0^2 c h^3) = 1.09737 \times 10^7$ m^{-1}. Significance: Directly predicts the wavelength of all spectral lines of single-electron systems (H, He^+, Li^{2+}) with astounding experimental precision. [1 Mark]

• **Key Exam Result:** 1 divided by $\lambda = R_H$ [1 divided by n_1^2 - 1 divided by n_2^2]; Rydberg constant $R_H = 1.097 \times 10^7$ m^{-1}

6. Numerical on shortest and longest wavelengths of Balmer and Lyman series [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** 1 divided by $\lambda = R$ [1 divided by n_1^2 - 1 divided by n_2^2], where $R = 1.097 \times 10^7$ m^{-1}. [1 Mark]

• **[Balmer Series Wavelengths ($n_1 = 2$)]** (a) Longest wavelength ($n_2 = 3$): 1 divided by λ max = R [one-fourth (0.25) - 1 divided by 9] = R (5 divided by 36) implies λ max = 36 divided by (5 R) = 36 divided by $(5 \times 1.097 \times 10^7) = 6.56 \times 10^{-7}$ m = 656.3 nm (Hα line). [1 Mark]

• (b) Shortest wavelength or series limit ($n_2 = \infty$): 1 divided by λ min = R [one-fourth (0.25) - 1 divided by ∞] = R divided by 4 implies λ min = 4 divided by R = 4 divided by $(1.097 \times 10^7) = 3.65 \times 10^{-7}$ m = 364.6 nm. [1 Mark]

• **Key Exam Result:** Balmer Series: λ max = 656.3 nm (Red); λ min = 364.6 nm (UV-Violet border)

7. Numerical on radius and energy of first Bohr orbit of He+ and Li++ ions [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Formulas for Hydrogen-like Ions]** Radius r n = $(a_0 n^2)$ divided by Z; Energy E n = - $(13.6 Z^2)$ divided by n^2 eV. Here $a_0 = 0.529$ Å. [1 Mark]

• **[Calculations for n = 1]** 1. Singly ionized Helium (He^+, Z = 2): $r_1 = (0.529 \times 1^2)$ divided by 2 = 0.2645 Å. $E_1 = - 13.6 \times (2)^2$ divided by 1^2 = - 13.6 × 4 = - 54.4 eV. [1 Mark]

• 2. Doubly ionized Lithium (Li^{2+}, Z = 3): $r_1 = (0.529 \times 1^2)$ divided by 3 = 0.1763 Å. $E_1 = - 13.6 \times (3)^2$ divided by 1^2 = - 13.6 × 9 = - 122.4 eV. [1 Mark]

• **Key Exam Result:** He^+: $r_1 = 0.265$ Å, $E_1 = -54.4$ eV | Li^{2+}: $r_1 = 0.176$ Å, $E_1 = -122.4$ eV